Topology Reference (Page 1/1)

Topology Brief: Mininet Routers + Switches + Hosts

Node Roles
r1, r2 : Linux routers (IP forwarding enabled)
s1, s2, s3 : L2 Open vSwitch switches
h1, h2 : User/client subnet
h3, h4 : Server subnet

Links
h1--s1, h2--s1, r1--s1
r1--s2--r2 (transit segment)
r2--s3, h3--s3, h4--s3

IP Plan
LAN A: 10.0.1.0/24 (gateway r1=10.0.1.1)
Transit: 10.0.12.0/30 (r1=10.0.12.1, r2=10.0.12.2)
LAN B: 10.0.2.0/24 (gateway r2=10.0.2.1)

Routing
r1 static route: 10.0.2.0/24 via 10.0.12.2
r2 static route: 10.0.1.0/24 via 10.0.12.1

Security-Relevant Control Points
1) Router FORWARD chain: east-west policy enforcement
2) Router INPUT chain: control-plane protection
3) Sysctl hardening knobs on routers

Implementation Actions Supported by Simulator API
- baseline hardening (sysctl + invalid-state drops)
- disable/block telnet forwarding (tcp/23)
- block ICMP from h1 to h3

Verification Signals
- Reachability probe h2->h4 for baseline health
- Policy probe h1->h3 ICMP expected fail when blocked
- iptables rule presence checks